



Mohamed Daoud

M.Sc. in Electrical Engineering & Sustainable Development

Contact

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Cairo - EGYPT

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My Portfolio

LANGUAGES

Français (C1) English (B2)

Arabic (C2 - Native)

TECHNICAL SKILLS

PROGRAMMING & AUTOMATION

Python C / C++ VHDL

Web (HTML5/CSS3)

Embedded (ESP32/Arduino)

FPGA / DSP / PIC

MBD, FEA & CAD DESIGN

MATLAB / Simulink Modelica LabVIEW

JMAG / FEMM COMSOL / Ansys

HEEDS Optimization SOLIDWORKS

CATIA / AutoCAD

POWER ELECTRONICS & PCB

PSIM LTspice PSpice Altium Designer

SYSTEM ENGINEERING

IBM DOORS Siemens Polarion

Jira / Confluence A-SPICE ISO 26262

IT, OPERATIONS & MANAGEMENT

Active Directory SAP ERP/CRM

Microsoft 365 Git / GitHub MS Project

EDUCATION

Master of Science in Engineering: Electrical Engineering & sustainable development

Université de Lille, école centrale Lille, école Arts et Métiers & L2EP laboratory

2018 – 2020

Lille, France

Bachelor of Science in Engineering: Electrical Engineering

Université de Lille - sciences et technologie

Major: Electronics, Electrical Energy and Control (EEA) | 2014 – 2018

Lille, France

PROFESSIONAL EXPERIENCE



Senior System Engineer: Sonnen GmbH

December 2025 – Present

New Cairo, Egypt

- MBSE:** Developed modular Multi-Physical simulation models for diverse residential battery storage System architectures: Battery modules, hybrid inverters, grid variants and PV systems for DC-coupled or AC-coupled retrofit designs.
- Simulation Platform:** Developed simulation platform with Model Exchange & cosimulation environments using FMI framework with single/multirate, adaptive/fixed-step configurations and implicit/explicit solvers
- Hardware Validation & SIL/HIL Testing:** Initiated validated multi-node battery and inverter model libraries, benchmarks, and model wrappers prepared for SIL/HIL loops, directly validating topologies against physical experimental test-bench data.



Senior Analyst: Capgemini

December 2024 – December 2025

New Cairo, Egypt

- Infrastructure Engineering & Incident Leadership:** Technical Lead delivering proactive IT infrastructure support and cross-team escalation handling for BAYER Crop Science, orchestrating strategic endpoint, network, and cloud maintenance to secure and optimize enterprise operations.



System Engineer: Technology & Strategy (T&S)

January 2024 – March 2024

Lyon, France

- ASPICE Traceability:** Managed full requirement traceability loops and controlled change verification workflows under automotive guidelines.
- Functional Safety:** Designed robust HW/SW system specifications aligned with safety objectives & Functional Safety under ISO 26262.



System Engineer: Valeo

April 2022 – January 2024

Gizeh, Egypt

- Automotive V-Model Execution:** Authored system element specs following strict A-SPICE guidelines and customer requirements.
- Cross-Functional Review:** Led design reviews with implementation and testing teams while owning risk and issue tracking.
- EV Powertrain Projects:** Coordinated deliverables for 48V Belt Starter Generator (iBSG) and 800V Silicon Carbide (SiC) Inverters.



Hardware Support Engineer: International Business Machines (IBM)

June 2021 – March 2022

Gizeh, Egypt

- Enterprise Server Diagnostics:** Ran advanced hardware troubleshooting on enterprise IBM/Lenovo System X and ThinkSystem lines.
- Infrastructure Deployment:** Managed complex machine installations, network configurations, and firmware lifecycle packages & Formulated immediate hardware action plans as the primary technical point-of-contact for corporates IT Admin.



System Design Engineer: Fluid Actuation & Control Toulouse (FACT Group)

April 2020 – October 2020

Toulouse, France

- Clean Sky project electrical valve design:** Modeled and sized an innovative electric linear actuator & Implemented strict mechanical/electrical boundary constraints to satisfy rigorous aviation safety standards.
- Multi-Variable Optimization:** Executed parametric optimization loops in HEEDS to minimize total aircraft valve system weight.



Internship: Laboratoire d'Electrotechnique et d'Electronique de Puissance (L2EP)

October 2019 – February 2020

Lille, France

- 3MW Wind Turbine Power system design and control:** In collaboration with L2EP power systems team, I contributed to the optimized design of a mechanical/electrical power conversion stages focusing on power electronics design and DFIM/PMSG control strategies, enhancing overall system efficiency.